

that the hypothesis generation stage of non-demonstrative inference is not subject to scientific theorizing, in contrast to the hypothesis evaluation stage.

The comprehension procedure has the effect of making sure that when Peter (the “pragmatic listener”) samples candidate interpretations, the interpretation satisfying his expectations is among the first few samples drawn, if not the first. In other words, Peter’s sampling of candidate interpretations is not based on a uniform draw, but is heavily skewed by the path of least effort. In order to complete our model of the hypothesis evaluation stage of the comprehension procedure/heuristic, we need to look closer at various properties of the comprehension heuristic.

4.2 Parallel adjustment of context, implicit import and explicit content

Relevance Theory emphasizes that comprehension is an on-line process:

These subtasks should not be seen as sequentially ordered: the hearer does not FIRST decode the logical form, THEN construct an explicature and select an appropriate context, and THEN derive a range of implicated conclusions. Comprehension is an on-line process, and hypotheses about explicatures, implicated premises and implicated conclusions are developed in parallel against a background of expectations which may be revised or elaborated as the utterance unfolds. (Wilson and Sperber 2004, p. 615)

Let us suppose that Peter’s language module has just parsed the partial sentence *I don’t want to drink...* and paired it with the propositional fragment MARY DOES NOT WANT TO DRINK_. Given the relevance expectation (c) in 1, this propositional fragment is enough for Peter that the manifestness of assumption (f) in 1 is raised enough to form the hypothesis that Mary could have intended to communicate this assumption. Next, Peter’s language module has parsed the next chunk of the sentence *I don’t want to drink energizing drinks...* and paired this chunk with the propositional fragment MARY DOES NOT WANT TO DRINK ENERGIZING DRINKS. This propositional fragment contains the concept ENERGIZING DRINK which we have seen gives Peter easy access to encyclopaedic information along the lines in (d) and (g) in 1. This encyclopaedic information, if accessed as contextual assumptions, makes transparent how the proposition (e) in 1 can provide logical (inferential) support to (f) in 1.⁸

Having unpacked the general idea of the parallel adjustment between implicit import, explicit content and contextual assumptions in on-line processing of utterances, it becomes clear that Wilson and Sperber’s rejection of the picture of utterance interpretation as a sequential process (from decoding to explicature to context selection and implicated conclusions) does not advocate for a model of comprehension as a process consisting of completely independent, encapsulated parallel processes. Rather, the idea is that the subtasks of comprehension are interconnected in the sense that early results of linguistic decoding may in the presence of strong expectations about cognitive effects give rise to early hypotheses about intended meaning which may later be substantiated by backwards inference.

This parallel adjustment of implicit import, explicit content and contextual assumptions can be modelled as interacting conditional inferences:

⁸Bergen, Levy and Goodman (2016) recognise the need for the parallel adjustment of explicit content and implicit import in their model of M-implicature comprehension. As the authors rightly remark, this makes their account close in spirit to relevance theory in this respect.

1. “Bottom-up” inference: from logical form via contextual assumptions and explicit content to implicatures (implicated conclusions).
 - For every logical form:
 - sample accessible developments of explicit content,
 - sample accessible contextual assumptions
 - yield a probability distribution over implicatures conditional on hypotheses about explicit meaning and contextual assumptions
2. “Top-down” inference: from accessible hypotheses about implicit conclusions via contextual assumptions to explicatures
 - For every accessible implicit conclusion, and every logical form:
 - sample accessible explicatures
 - sample accessible contextual assumptions
 - yield a probability distribution over explicatures conditional on hypotheses about implicatures and contextual assumptions

In this model, the theorist/experimenter supplies a small set of hypothesis triplets <expl, cont, impl> ordered according to the assumed relative order of accessibility. Sampling is done over this ordered set. By changing the content of the hypotheses, as well as by changing the order of relative accessibility, the experimenter can check the results of this inference model against theoretical expectations.

Considering our main example, we propose to operate with the following simplifications:

- While every utterance could in principle have several logical forms if the utterance involves structural or lexical ambiguity, this issue does not arise in the example. Therefore we abstract away from this possibility.
- While the development of logical forms may involve free enrichment processes as well as filling in of variable values and ad-hoc concept construction for almost every word in the utterance, the example was chosen deliberately with the purpose to limit the amount of logical form enrichment. Therefore we can justifiably limit our attention to a narrow subset of factors. We consider only the possible reference resolution of “energizing drinks”, and possible on temporal interpretation.

The practical implications of the claim that implicit import and explicit content is adjusted in parallel can be seen more clearly in example (18b) from Wilson and Sperber (2002) (reprinted in Wilson and Sperber 2012, pp. 47–83.).

- (18) a. ALAN JONES: Do you want to join us for supper?
 b. LISA: No, thanks. I’ve eaten. (Wilson and Sperber 2012, p. 77)

Lisa’s utterance *I’ve eaten* can be understood as a legitimate reason to reject an invitation for supper so long as the temporal reference can be inferentially determined to refer to sometime this evening, and the concept denoted by the verb *eaten* be narrowed to convey the concept EAT SUPPER giving the idea that Lisa has eaten something of a kind an quantity that it counts as a full meal. Only when the logical form of Lisa’s utterance is developed in this way does the resulting explicature warrant the inference (19).

- (19) a. *Implicit premise*: If someone has already eaten a satisfying meal recently then that person has a good reason to reject the offer of a meal
 b. *Explicit premise*: Lisa has already eaten supper this evening
 c. *Implicit conclusion*: Lisa has a good reason to reject an offer of a meal

In fact, Wilson and Sperber go on to show that explicatures typically require a good deal of inferential adjustment to support implicatures to reach an interpretation that satisfies relevance expectations, a point further developed in Robyn Carston (2002). However, we have purposefully chosen an example where the amount of inferential adjustment to the explicature is arguably minimal, in order to focus more clearly on the implicatures.

What these considerations show is that there is no need for a stage in processing where a representation that is more developed than the logical form of the utterance, but less developed than the explicature (the intended proposition) plays a role. In other words, there is no level like the Gricean concept of *What is said* or a *Literal meaning* of the utterance that plays a role in utterance interpretation even though it may need to be adjusted, or even rejected and replaced, in later stages of interpretation. So when we develop a probabilistic model of inference in utterance comprehension, we must make sure that neither the speaker nor the listener is reasoning about a “literal listener” or a “literal speaker”, as it is standard practice in RSA work.

4.3 The central role of relevance expectations and their fine-tuning

4.3.1 Relevance expectations, strong and weak implicatures

Recall that the stopping criterion of the comprehension heuristic makes reference to the audience’s expectations of relevance. Expectations of relevance are raised in general terms by the communicative principle of relevance. However, for this general presumption to be operative in the comprehension heuristic, it must be fine-tuned with reference to the audience’s actual expectations at the particular moment. This is how (c) differs from (b) in table 1.

The example we are discussing involves an indirect answer to a specific question. In this case the audience has a very specific expectation of cognitive effects, indeed: that the communicator will convey some proposition that either affirms the propositional content of the yes-no question (by contextual strengthening), or negates it (by contextual contradiction and elimination). It is this very specific expectation that drives the parallel adjustment of explicit content, implicit import and context by backwards inference.

However, not every utterance raises specific expectations of relevance. Consider (20): Mary is walking in town and a passer-by asks her the time.

- (20) a. *Passer-by*: Excuse me, what’s the time?
 b. *Mary*: It’s 2:30 pm.
 c. [Proposition conveyed by Mary:] THE TIME AT PRESENT IS 2:30 PM

What is the passer-by’s expectation of relevance? Mary has no way to know, other than that a true answer (to the best of Mary’s capabilities) will be optimally relevant to him. Mary can be sure that when she communicates a true proposition along the